

SE LAB 5

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Issue Table:

Issue	Tool	Original Line(s)	Description	Fix Implemented
1.Mutable default arg	Bug	7	Logs= [] shared across calls.	Changed logs= [] to logs=None and initialized logs = [] inside the function body.
2.Missing module docstring	Pylint (C0114)	1	The main file is missing a docstring describing its purpose.	Added a descriptive module docstring at the top of the file.
3.Invalid function names	Pylint (C0103)	All function definitions	Function names like addItem and removeItem do not conform to the snake_case naming convention (e.g., add_item).	All functions renamed to use snake_case (e.g., add_item, load_data).
4.Missing function docstrings	Pylint (C0116)	All function definitions	Several functions are missing docstrings, making the code harder to understand.	Docstrings added to every function.
5.Unused import (logging)	Pylint (W0611) / Flake8 (F401)	Import section, Line 2	The logging module is imported but never used in the script.	Removed the import logging statement from the file.
6.Bare except / try-except-pass	Pylint (W0702) / Flake8 (E722) / Bandit (B110)	remove_item function	The code uses a bare except statement, which catches all exceptions, including KeyboardInterrupt. It can also hide programming errors.	Replaced the bare except with a specific except KeyError block for safe handling.
7.Use of eval ()	Bandit (B307) / Pylint (W0123)	main function (Line 58 in original)	The eval () function is used, which is a significant security vulnerability as it can execute arbitrary code from untrusted input.	The dangerous eval () call was removed entirely from main ().
8.Unspecified encoding	Pylint (W1514)	load_data, save_data	The open() calls on these lines do not explicitly specify a character encoding, which can lead to platform-specific errors.	Added explicit encoding="utf-8" argument to all open () calls.
9.Missing 'with' statement	Pylint (R1732)	load_data, save_data	The open() calls are not within a with statement, which means the file might not be properly closed in all cases.	Used the with open(...) construct for proper resource management.

10.Expected blank lines	2	Flake8 (E302, E305)	Throughout the file	Function definitions are not separated by two blank lines as required by PEP 8.	Ensured two blank lines separate all top-level function definitions.
11.Could be an f-string		Pylint (C0209)	add_item function	The use of format() for string formatting is less readable than an f-string.	Replaced old string formatting with a modern f-string.
12.TypeError (runtime error)		Runtime Issue	add_item (when called from main)	add_item (when called from main)	Added input validation to add_item to check if qty is numeric, preventing the error caused by "ten".
13.Using global statement (W0603)		Pylint (W0603)	load_data	The global keyword is used, which can make it difficult to track changes to variables.	The global keyword is still used in load_data and implicitly in add_item and remove_item (due to modifying stock_data), but it is necessary here to modify the top-level dictionary, making it an acceptable trade-off for this style of script.

